1) Solve the system of equatioı by using the addition method.

$x + y = 10$ equation (1)

$x - y = 6$ equation (2)

a) Add equation (1) and equation (2). Result: _____ ? _____

b) Use the result from part (a) to solve for x. $x =$ _____ ? _____

c) Use equation (1) to solve for y. $y =$ _____ ? _____

2) Solve the system of equations by using the addition method.

$x + 2y = 12$ equation (1)

$x - 2y = 16$ equation (2)

a) Add equation (1) and equation (2). Result: _____ ? _____

b) Use the result from part (a) to solve for x. $x =$ _____ ? _____

c) Use equation (1) to solve for y. $y =$ _____ ? _____

3) Solve the system of equations by using the addition method.

$-3x + y = 8$ equation (1)

$3x - y = 4$ equation (2)

a) Add equation (1) and equation (2). Result: _____ ? _____

b) Use the result from part (a) to solve for x. $x =$ _____ ? _____

c) Use equation (1) to solve for y. $y =$ _____ ? _____

4) Solve the system of equations by using the addition method.

$4x + y = 5$ equation (1)

$4x + y = 7$ equation (2)

a) Multiply equation (1) by -1. Result: _____ ? _____

b) Add result of part (a) and equation (2). Result: _____ ? _____

c) Use the result from part (b) to solve for x. $x =$ _____ ? _____

d) Use equation (1) to solve for y. $y =$ _____ ? _____

5) Solve the system of equations by using the addition method.

$3x + 4y = 12$ equation (1)

$4x - 5y = 16$ equation (2)

a) Multiply equation (1) by 5. Result: _____ ? _____

b) Multiply equation (2) by 4. Result: _____ ? _____

c) Add result of part (a) and result of part (b). Result: _____ ? _____

d) Use the result from part (c) to solve for x. $x =$ _____ ? _____

e) Use equation (1) to solve for y. $y =$ _____ ? _____

6) Solve the system of equations by using the addition method

$3x + 5y = 8$ equation (1)

$-2x + 3y = 7$ equation (2)

a) Multiply equation (1) by 2. Result: _____?_____

b) Multiply equation (2) by 3. Result: _____?_____

c) Add result of part (a) and result of part (b). Result: _____?_____

d) Use the result from part (c) to solve for y. $y =$ _____?_____

e) Use equation (1) to solve for x. $x =$ _____?_____

7) Solve the system of equations by using the addition method

$2x + 3y = 1$ equation (1)

$-4x - 6y = 7$ equation (2)

a) Multiply equation (1) by 2. Result: _____?_____

b) Add result of part (a) and equation (2). Result: _____?_____

c) Use the result from part (b) to solve for x and y. $x =$ ___?___ , $y =$ ___?___

8) Solve the system of equations by using the addition method

$4x + 3y = 5$ equation (1)

$-8x - 6y = -10$ equation (2)

a) Multiply equation (1) by 2. Result: _____ ? _____

b) Add result of part (a) and equation (2). Result: _____ ? _____

c) Use the result from part (b) to solve for x and y. $x =$ ___ ? ___ , $y =$ ___ ? ___

9) Company ABC sells hamburgers and sodas at a baseball game.
Each hamburger costs \$7.50 and each soda costs \$1.50.
At the end of the day, Company ABC made \$945.
Company ABC sold a total of 150 hamburgers and sodas combined.
How many hamburgers and sodas were sold?

Let x be the number of hamburgers sold

Let y be the number of sodas sold

a) $7.50x + 1.50y =$ ___ ? ___

b) $x + y =$ ___ ? ___

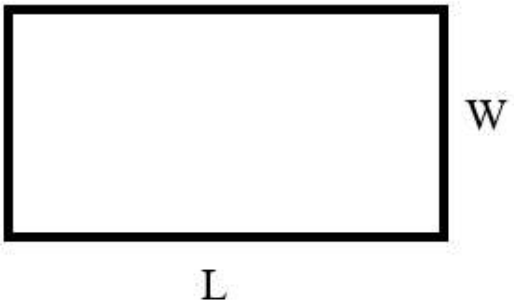
c) Multiply equation in part (b) by -1.50. Result: _____ ? _____

d) Add result of part (c) and part (a). Result: _____ ? _____

e) Use the result from part (d) to solve for x. $x =$ ___ ? ___

f) Use the equation from part (a) to solve for y. $y =$ ___ ? ___

10) The length of a rectangular parking plot is 125 feet longer than the width. If the perimeter is 3000 feet, find the dimensions of the parking lot.



Let L be the length of the parking lot.

Let W be the width of the parking lot.

a) Express length L in terms of width (W). $L =$ ___?___

b) $2L + 2W =$ ___?___

c) From part (a), $L - W =$ ___?___

d) Multiply equation in part (c) by 2. Result: ___?___

e) Add equation in part (b) and result from part (d). Result: ___?___

f) Use the result from part (e) to solve for L. $L =$ ___?___

g) Use the equation from part (a) to solve for W. $W =$ ___?___

11) John invested \$12,000 in bonds and money market funds. He invested a portion of the money in bonds paying 7% interest per year and the remainder in a money market fund paying 6% per year. After one year the total income he had earned from the investments was \$800. How much had John invested at each rate?

Let B be the amount invested in bonds.

Let M be the amount invested in money market funds.

a) $B + M =$ ___?___

b) $0.07B + 0.06M =$ ___?___

c) Multiply equation in part (a) by -0.07. Result: ___?___

d) Add equation in part (b) and result from part (c). Result: ___?___

e) Use the result from part (d) to solve for M. $M =$ ___?___

f) Use the equation from part (a) to solve for B. $B =$ ___?___

12) Tickets for a baseball game were \$5.50 for general admission and \$2.50 for kids. If there were 5 times as many general admission tickets sold as there were kids' tickets sold, and total receipts were \$8,400, how many of each type of ticket were sold?

Let G be the number of general admission tickets sold.

Let K be the number of kids' tickets sold.

a) Express G in terms of K. G = ___?___

b) 5.50G + 2.50K = ___?___

c) From part (a), G − ___?___ = 0

d) Multiply equation in part (b) by 2. Result: ___?___

e) Add equation in part (c) and result from part (d). Result: ___?___

f) Use the result from part (e) to solve for G. G = ___?___

g) Use the equation from part (a) to solve for K. K = ___?___